Which SQL keyword is used for removing duplicates from the result of a query statement?

- ○ Distinct
- ○ All
- ○ Any
- ○ Exists

Consider a table A with 30 rows and a table B with 2 rows. How many rows will be there in the Cartesian Product of A and B? ***1 point***

- ○ 30
- ○ 50
- ○ 60
- ○ 2

Identify the appropriate SQL statement that finds the names of the employees whose age is not greater than 30 and whose address is 'Seattle'. ***1 point***

○
```
SELECT name
FROM employee
WHERE age> 30 OR address= 'Seattle' ;
```

○
```
SELECT name
FROM employee
WHERE age<=30 AND address= 'Seattle' ;
```

○
```
SELECT name
FROM employee
WHERE age=30 AND address= 'Seattle' ;
```

○
```
SELECT name
FROM employee
WHERE age>30 OR address= 'Seattle' ;
```

Given that a table named **employee** has an attribute named age with a varchar data type which stores the age of an employee. Which among the following ages can be returned by the given query? ***1 point***

SELECT * FROM employee WHERE age like '%1';

- ☐ 11

☐ 211

☐ 312

☐ None of the above

Consider the given queries: *1 point*

1. SELECT * FROM employee order by salary asc;
2. SELECT * FROM employee order by salary ;

Choose the correct option.

◯ Query 1 displays the salary of all the employees, Query 2 has errors in it.

◯ Query 1 displays the salary of all the employees in ascending order, Query 2 displays the salary of the employees randomly.

◯ Both queries produce the same output.

◯ Query 1 displays the salary of all the employees in ascending order, Query 2 displays the salary of the employees in descending order.

Union is a .... set. *1 point*

◯ Pure

◯ Multi

◯ Both

◯ None of the above

Consider the table **department** (dept_name, dept_id, city). Identify the correct statement(s) that find(s) the name of the departments where the city is either ‘Paris’ or ‘Mexico’. *1 point*

☐ Select dept_name from department where city=‘Paris’
UNION
Select dept_name from department where city=‘Mexico’;

☐ Select dept_name from department where city=‘Paris’
EXCEPT
Select dept_name from department where city=‘Mexico’;

Select dept_name from department where city='Paris'
INTERSECT
☐ Select dept_name from department where city='Mexico';

Select dept_name from department where city='Paris'
UNION ALL
☐ Select dept_name from department where city='Mexico';

Which of the following aggregate functions returns the number of rows that matches the required specification? ***1 point***

○ sum()

○ count()

○ avg()

○ max()

Consider two relational schemas as follows: ***2 points***

student(id_no,name,dept_name)
department(dept_name,budget,dept_no)

Find the list of students of those departments which have budget greater than 1000.

○
```
select name, budget
from student e, department d
where e.dept_name = d.dept_name
and budget>1000;
```

○
```
select name, budget
from student e, department d
where e.dept_name = d.dept_name
budget=>1000;
```

○
```
select name, budget
from student e, department d
where e.dept_name = d.dept_name
budget<1000;
```

○
```
select name, budget
from student e, department d
where e.dept_name = d.dept_name
budget=1000;
```

Which of the following joins is similar to Cartesian product? ***1 point***

◯ Theta join

◯ Left outer join

◯ Cross join

◯ Natural join

Consider two relational schemas as follows: ***1 point***

employee(emp_num,emp_name,salary,dept_num)
department(dept_num,dept_name)

Suppose **employee** table has 5 tuples and 6 attributes, and the department table has 0 tuples and 6 attributes. When a CROSS JOIN is performed between **employee** and **department**, how many tuples are there?

◯ 12

◯ 96

◯ 56

◯ 0

Given relation ***1 point***

student(id,name,age,ph_no)

Find the names of all students whose name ends with ”gar”.

◯ select name from student where name like ’%gar%’;

◯ select name from student where name like ’%gar’;

◯ select name from student where name like ’gar%’;

◯ select name from student where name like ’gar’;

Consider the following statements: ***2 points***

UNION removes duplicate rows.
UNION ALL removes duplicate rows.

◯ Both statements are correct

◯ Both statements are wrong

◯ Statement 1 is wrong, statement 2 is correct

◯ Statement 2 is wrong, statement 1 is correct

What will be the result of the following query? **2 points**

```
select name
from instructor
where dept_name in('Music','Biology')
intersect
select name
from instructor
where salary>6000;
```

◯ The names of all instructors who taught in either Music department or the Biology department and whose salary is greater than 6000

◯ The names of all instructors who taught in both Music department and the Biology department and whose salary is less than 6000

◯ The names of all instructors who taught in either Music department and the Biology department and whose salary is greater than 6000

◯ The names of all instructors who taught in either Music department or the Biology department or whose salary is greater than 6000

Which of the following is/are aggregate function(s)? **1 point**

☐ min

☐ avg

☐ view

☐ with

Among the given options, what will be the result of the following query? **1 point**

```
select building, avg(capacity)
from classroom
group by building
having avg(capacity)<30;
```

◯ The names and the average capacity of each building whose average capacity is less than 30

- The names and the average capacity of each building whose average capacity is equal to 30
- The names and the average capacity of each building whose average capacity is less than or equal to 30
- The names and capacity of each building whose average capacity is greater than 30